

Ayman Patel

London, UK | ayman.patel97@gmail.com | linkedin.com/in/aymanapatel
github.com/aymanapatel | patelofthought.com/me | patelofthought.com/blog

SUMMARY

Full-Stack Engineer with 6+ years at Mastercard, targeting privacy-first product engineering with a frontend focus across React, TypeScript/JavaScript, Redux Toolkit, RTK Query, Vite and Webpack. I designed a microfrontend platform that let 10+ teams deploy independently, cut UI load time by 30%, and built security-sensitive payment services backed by Java, Spring, PostgreSQL, Redis and Kafka. I take work from product requirements and Figma through implementation, automated testing, CI/CD, observability and production support. Backend experience includes Node.js, NestJS, Go, gRPC, REST APIs, MongoDB, RabbitMQ, AWS, Terraform and Kubernetes.

EDUCATION

Queen Mary University of London

MSc Advanced Computer Science

September 2025 - September 2026

Sardar Vallabhbhai Institute of Technology

Bachelor of Computer Science, GPA: 7.9/10

July 2015 - May 2019

EXPERIENCE

Mastercard

Senior Software Engineer

March 2024 - August 2025

Vadodara, India

- Designed and rolled out a microfrontend platform with Webpack Module Federation, splitting a monolithic React frontend into independently deployable modules. It enabled **10+ product teams to ship on separate release cadences** and cut deployment coordination **from days to hours**.
- Built the host shell with React, TypeScript, Redux Toolkit and RTK Query. Defined shared dependency boundaries, runtime module discovery and versioning contracts so teams could upgrade libraries without breaking the combined product.
- Built **customer onboarding features with NestJS and Node.js** at Mastercard.
- Led the accessibility initiative across React and TypeScript products, lifting WCAG compliance by **40%**, coaching **10+ engineering teams** on semantic markup, focus management and ARIA patterns, and contributing 5 fixes to Mastercard's internal design system.
- Cut UI load time by **30%** through Akamai CDN migration and bundle delivery work. Added client-side traces and Core Web Vitals with Grafana and OpenTelemetry so regressions were visible before production.
- Built Spring, Spring Data JPA and Hibernate APIs with PostgreSQL JSONB, Redis and Kafka. Added circuit breaker and retry patterns across 5+ microservices, improving system availability by **15%**.

Mastercard

Software Engineer II

March 2022 - February 2024

Vadodara, India

- Built greenfield React and TypeScript applications with Redux and RTK Query, plus Java and Spring backend services with Oracle and Hibernate.
- Wrote UI and component tests with Jest and React Testing Library, added **snapshot tests for Storybook components**, and covered end-to-end flows with Playwright and Cypress. Reached **95% test coverage** and cut the CI end-to-end suite **from 8 hours to 3 hours** through splitting, parallel execution, selective retries and environment caching.
- Produced high-fidelity Figma prototypes with product managers and stakeholders, setting the UX direction in **2 days** when no UX designer was available.

Mastercard

Software Engineer I

July 2019 - February 2022

Vadodara, India

- Designed REST APIs for Camunda workflows and maintained a Hardware Security Module integration for MDES tokenisation, **securing card data transmission across mobile payment flows**.
- Reviewed penetration-test findings, prioritised fixes by risk and shipped remediation to keep enterprise applications compliant.

Mastercard

Software Engineering Intern

January 2019 - June 2019

Vadodara, India

- Built a non-blocking Spring WebFlux image service with Reactive Streams and ImageMagick compression, using TDD to protect behaviour and support safe refactoring.

HACKATHON

- **{Tech: Europe} Hackathon, London, 2026 | Most Secure Build Award (EUR 1,000):** Built a deep-science research product that turned academic papers into investor-ready intelligence. Aikido Security found **zero issues in the codebase**. Partners included Attio, Tavily and Google DeepMind.
- **Technology Hackathon, 2023 | 2nd place:** Migrated a React application to Next.js, improving page load performance and adding server-side rendering and static generation for key routes.
- **Encode Hackathon, London, 2025 | 3rd place:** Built Phantom Fraud, a Solana risk engine that parsed on-chain transactions and detected wash-trading patterns through circular wallet flows and buyer-diversity heuristics.

BLOG

- **Patel of Thought | 40+ technical articles:** I write about production tradeoffs across frontend architecture, observability, distributed data infrastructure and AI/LLM systems. The work has appeared in industry newsletters, including eBPF.
- **Snapshot testing:** Published a practical guide to snapshot testing for frontend components.
- **Frontend engineering whitepapers:** Wrote company-wide papers on accessibility and microfrontend architecture. They earned CTO recognition, became internal engineering references and reached **200 engineers**.

PROJECTS

Web Accessibility Repair Platform | React, FastAPI, Playwright, axe-core

Built an end-to-end platform that collected HTML, accessibility-tree, visual and screenshot data from 704 public and authenticated webpages, then generated schema-constrained repairs through FastAPI and React. Improved page-level F1 by **34.1 points** and rule-level F1 by **30.6 points**.

Snapshot Testing | Storybook, component testing

Built snapshot tests for Storybook components and wrote a companion guide explaining how stored component output catches unexpected UI changes.

Go Microservices | Go, REST, gRPC, Kubernetes

Built a production-oriented Go microservices platform with REST and gRPC endpoints, Gorilla Mux request validation, Swagger/OpenAPI contracts, Envoy service mesh and load balancing, plus Kubernetes/WASM filters.

FELLOWSHIP

Nebius Academy

AI Engineering Fellow

March 2026 - May 2026

London, UK

- **LLM architectures:** Learned how pre-norm, causal attention, LoRA, MoE, RoPE and KV caches shape transformer training and inference. Built a transformer from scratch and LoRA for GPT-2, training 811K of 124M parameters.
- **AI agents:** Learned where exploratory agents need deterministic rules and traceable tool calls. Built a LangGraph ReAct agent, Rasa CALM booking flow and shared MCP tool layer.
- **MLOps and distributed training:** Learned to debug distributed jobs across model code, containers and NCCL networking. Built a version-controlled DDP pipeline across 2 Nvidia H100 Kubernetes nodes with Docker, SkyPilot and torchrun.
- **GPU performance:** Learned to use profiles and roofline analysis to find memory-bound kernels. Built KV caching and preallocated buffers, cutting 128-token inference **from 0.943s to 0.141s**.